Lesson notes · for you

# Loss, descent and backprop

University, 2

**Time**
45 min

**Materials**
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

## Learning goals

- Loss is large when wrong
- Step against the slope

## Activity: In the lab

1. Run Good
2. Run Too big

## Discussion

- Where does the gradient point?

## Check yourself

- $\theta^2$ at 3, rate 0.1: write the next θ.

**Try it this week**
Compute the 2.4 step on paper without looking it up.

Worksheet

# Loss, descent and backprop

University, 2

Name _____ Class _____ Date _____

**1** **Loss $\theta^2$, $\theta = 3$, $\eta = 0.1$. $\theta$ after one step?**

◯ 2.4 ◯ 2.7 ◯ 3.6

**2** **The gradient points at the minimum.**

◯ True ◯ False

**3** **Match the curve to the cause.**

| | | |
|---|---|---|
| Loss grows every step | • • • • | Memorising the training set |
| Loss falls very slowly | • • • • | Step too small |
| Train loss low, new data worse | • • • • | Step too big |

**4** **Order one descent step.**

☐ Step against the gradient

☐ Compute the gradient

☐ Compute the loss

✎ **What does a growing loss tell you, before you look at the weights?**